NEUROVOX — Viva Questions & Answers

Short answers in simple words · say them out loud once or twice

A. About the Project

Q1. What is your project in one line?

Neurovox is a Java app that listens to a 5-second "aaah" sound, measures how shaky the voice is, and tells whether the person is at **low or high risk** of Parkinson's disease.

Q2. What is Parkinson's disease?

It is a brain disease where the cells that make **dopamine** (a chemical that controls movement) slowly die. This causes shaking, stiffness, slow movement and **voice changes**.

Q3. Why use the voice?

Up to **90% of patients** get voice problems, often early. Speaking needs very fine muscle control, so small damage shows up in the voice first. And a voice test needs only a microphone, so it is free and painless.

Q4. Why a sustained "aaah" and not normal speech?

"aaah" is a **steady sound**, so the wave repeats nicely and is easy to measure. It also has no words, so it works for **any language**.

Q5. Is this a diagnosis?

No. It is only a **screening** tool. It says low risk or high risk, so the person knows whether to see a neurologist. Only a doctor can diagnose.

Q6. What was your part in the project?

I handled the **signal processing**: cleaning the audio (even volume, removing silence, cutting into frames) and **finding the pitch** with autocorrelation, which we compared with the YIN method.

Tip: Change this if the examiner asks a teammate. Bhanuteja: features and ML. Md. Ali: the app, server, database and website.

B. The Voice Features

Q7. What is F0 (pitch)?

F0 is the **fundamental frequency**, meaning how many times the vocal cords vibrate per second. Formula: **F0 = sample rate / pitch period**. Unit: Hz. Example: 44100 / 302 = 146 Hz.

Tip: Normal: men about 85–180 Hz, women about 165–255 Hz.

Q8. What is jitter?

Jitter is how much the **pitch shakes** from one vibration to the next. We take the average jump between neighbouring periods and divide by the average period, times 100. Healthy is below about **1%**.

Q9. What is shimmer?

Shimmer is how much the **loudness shakes** from one vibration to the next. It is the same formula as jitter, but on amplitudes (wave heights). Healthy is below about **3.8%**.

Q10. Difference between jitter and shimmer?

Jitter is shaking in **time** (pitch). Shimmer is shaking in **height** (loudness). Example: clapping where the gaps keep changing is jitter; clapping where some claps are loud and some soft is shimmer.

Q11. What is HNR?

Harmonics-to-Noise Ratio: how much of the voice is clean tone compared with breathy noise. Unit: dB. A **higher HNR means a cleaner voice**. Healthy is usually above about 20 dB.

Q12. Why do these show Parkinson's?

Parkinson's weakens the muscles around the vocal cords. So the vibration becomes uneven (**jitter goes up**), the loudness becomes uneven (**shimmer goes up**) and the voice gets breathy (**HNR goes down**).

Q13. What is the pitch period? What is 302?

The pitch period is the length of one vibration, counted in samples. 302 means the wave repeats every 302 samples. In the full app, **findPeriod()** finds it from the recording; in the exam code we typed it in.

Q14. What is the sample rate 44100?

The microphone takes **44,100 readings per second**. It is the standard CD-quality rate.

C. Algorithms and Data

Q15. What is autocorrelation?

We compare the sound with a **shifted copy of itself**. When the shift equals one vibration, the two copies line up best and give the biggest sum. That shift is the pitch period.

Tip: Analogy: sliding two copies of a striped paper over each other; they match perfectly when you shift by exactly one stripe.

Q16. Why minLag and maxLag?

To search only in the **human voice range** (about 75–500 Hz). For example, sampleRate/500 to sampleRate/75. This makes it faster and avoids wrong answers.

Q17. What is YIN?

An improved version of autocorrelation that uses the **difference** between the sound and its shifted copy. It makes fewer mistakes like picking double or half the real pitch. We compared it with our autocorrelation.

Q18. What machine learning model did you use, and why?

Logistic regression. It is simple, works well on a small dataset, gives a **probability** (0 to 1), and its weights show which feature matters most, so it is easy to explain.

Q19. How does logistic regression work?

1) Make a score: **z = bias + w1·jitter + w2·shimmer + w3·HNR**. 2) Pass it through the **sigmoid**: P = 1 / (1 + e^{-z}). 3) If P >= 0.5, high risk; else low risk.

Q20. What is the sigmoid function?

A function that squeezes **any number into the range 0 to 1**. A big positive z gives nearly 1, a big negative z gives nearly 0, and z = 0 gives exactly 0.5. So we can read it as a probability.

Q21. Why is the HNR weight negative?

Because a **higher HNR means a healthier voice**, so it should **lower** the risk. Jitter and shimmer have positive weights because more shaking means more risk.

Q22. Why the 0.5 threshold?

It is the middle point. For screening we could **lower it** (say 0.4) to catch more possible patients, since missing a sick person is worse than a false alarm.

Q23. Which dataset did you use?

The **UCI Oxford Parkinson's Dataset**: 195 voice recordings from 31 people (23 with Parkinson's), with features like jitter, shimmer and HNR.

Q24. What is data leakage? How did you avoid it?

Each person has several recordings. If the same person is in both training and testing, the model learns to **recognise the person**, not the disease, so the accuracy looks falsely high. We split **by patient**, so the test people are completely new.

Q25. What is Praat and why did you use it?

Praat is a free, standard **voice-analysis tool** used by scientists. We checked our Java jitter, shimmer, HNR and F0 values against Praat to prove our code gives correct numbers.

Q26. What accuracy did you get?

Give your team's real number from your results slide.

Tip: Don't make up a number. If you're unsure, say: "We tested on new patients only, so our accuracy is honest."

D. Java Code Questions

Q27. What does FeatureExtractor.pitch(44100, 302) mean?

It calls the **pitch** method inside the **FeatureExtractor** class, sends 44100 and 302 as inputs, and stores the returned answer (146.0) in f0.

Q28. Why are the methods static?

They only do calculations and don't need any stored data, so we call them with the **class name** and don't need an object (like Math.sqrt()).

Q29. Why two classes?

To keep each job separate. **Main** runs the flow and makes the prediction; **FeatureExtractor** does the voice math. The code is cleaner, reusable and easy to test. This is **abstraction**: Main uses the methods without knowing the math inside.

Q30. Why (double) in sampleRate / period?

Both are int, and int / int drops the decimal (44100 / 302 = 146). Casting to double keeps the decimal (146.03).

Q31. Why x.length - 1 in variation()?

5 values have only **4 gaps** between neighbours, so we divide the total jump by n – 1.

Q32. Why Math.abs()?

A jump can go up or down. abs() makes it positive, so up and down jumps don't cancel each other out.

Q33. What is for (double v : x)?

An **enhanced for loop** (for-each). It visits every value in the array one by one, here to add them all up.

Q34. What do %.2f, %% and %n mean in printf?

%.2f prints a decimal with 2 digits. **%%** prints the % symbol itself. **%n** starts a new line.

Q35. What does Math.exp(-z) do?

It calculates $\mathbf{e^{-z}}$ ($e \approx 2.718$), which is needed for the sigmoid function.

Q36. Why can only one class be public in a file?

Java's rule: a public class must have the **same name as the file**. Main.java can only have public class Main, so FeatureExtractor is either non-public or in its own file.

Q37. Which OOP concepts did you use?

Classes and objects (Patient, VoiceSample), **encapsulation** (data kept private with methods to use it), **abstraction** (FeatureExtractor hides the math), and **packages** (model, audio, ml, database, ui).

Q38. How does the frontend talk to Java?

Through Java's built-in **HttpServer** (JDK). The browser sends the recording over HTTP, Java processes it and sends the result back. We used **SQLite** through **JDBC** to save results.

E. Limits and Future

Q39. What are the limitations?

Small dataset (31 people). Background noise and a bad mic can change the results. Other problems like a cold or throat infection can also make the voice shaky. And it is only screening, not diagnosis.

Q40. Future scope?

More features (like MFCC), a bigger and more varied dataset (including Indian voices), a mobile app, and tracking a patient's voice over months.